

Sociotechnical Safety Evaluation of Generative AI Systems

Summary

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This paper, authored by Laura Weidinger and colleagues at Google DeepMind, addresses the critical need for comprehensive safety evaluation of generative AI systems. The authors argue that current evaluation approaches are heterogeneous, ad hoc, and overwhelmingly focused on technical capabilities alone, thereby failing to account for the social and contextual factors that determine whether a given AI capability actually causes harm. The paper makes two principal contributions: a three-layered sociotechnical framework for structuring safety evaluations, and an empirical survey of the existing evaluation landscape that reveals significant gaps.

The proposed framework comprises three layers that progressively incorporate broader context. The first layer, **“capability evaluation”**, targets technical components and AI system outputs in isolation—assessing, for example, whether a model produces factually incorrect or stereotypical content. The second layer, **“human interaction evaluation”**, centers the experience of people using AI systems, examining how context, user characteristics, and application design influence whether capabilities translate into actual harm. This includes studying phenomena such as overreliance, persuasion effects, and disparate impacts across user groups. The third layer, **“systemic impact evaluation”**, assesses broader societal, economic, and environmental effects that emerge when AI systems operate at scale, including labor market disruptions, erosion of public goods, and environmental costs of computational infrastructure.

The authors' survey of existing evaluations, compiled into a publicly released repository, reveals three critical gaps. First, a **“coverage gap”**: many identified risk areas—particularly information hazards, threats to human autonomy, and socioeconomic harms—lack adequate evaluations. Second, a **“context gap”**: the vast majority of existing evaluations operate at the capability layer, with human interaction and systemic impact evaluations being rare. Third, a **“multimodal gap”**: approximately 75% of evaluations target text output exclusively, despite growing deployment of image, audio, and video generative systems that present distinct and compositional risks across modalities.

To address these gaps, the paper outlines practical steps including careful operationalization of complex harm constructs into measurable metrics at each evaluation layer, repurposing text-based evaluations for other modalities, and leveraging model-driven evaluation approaches. The authors also delineate roles and responsibilities across stakeholders—AI

developers, policymakers, civil society, and researchers—emphasizing that evaluations must become standardized, routine, and consequential rather than optional exercises. They candidly acknowledge that evaluation is inherently incomplete and value-laden, as normative choices pervade decisions about what to measure and what constitutes acceptable performance.

The broader significance of this work lies in its argument that safety evaluation must evolve from component-level technical testing toward systems thinking that integrates human and societal dimensions. By providing a structured yet flexible framework, the authors offer a tractable path toward more robust, comparable, and comprehensive safety assurance for generative AI systems at a time when their societal integration is rapidly accelerating.